

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Large contracted project portfolios expose governing boards to a recurring operational decision absent from the formal literature: how to allocate budget across active projects at each discrete period to maximise discounted net cash flow under cash-flow uncertainty, FIDIC contract mechanics, and the options to intervene in or terminate underperforming projects. This paper addresses that gap through a three-level modelling arc. **Level 1** formulates the problem as a deterministic full-foresight Mixed-Integer Linear Program whose optimum constitutes a provable upper bound on any causal policy operating under uncertainty. **Level 2** reintroduces stochastic contractor performance, establishing that the resulting multi-stage stochastic program is computationally intractable at realistic portfolio scales due to exponential scenario-tree growth. **Level 3** reformulates the problem as a Constrained Markov Decision Process, implemented as a configurable Gymnasium-compatible environment calibrated from anonymised EPC contractor data, on which a Proximal Policy Optimisation agent is trained and evaluated against the Level 1 oracle, a two-stage stochastic program, and a rule-based heuristic. The environment itself—encoding advance payment recovery, milestone-based payments, retention release, symmetric deviation penalties, and a two-condition termination gate—constitutes the primary contribution.

Keywords: project portfolio management; budget allocation; reinforcement learning; constrained Markov decision process; earned value management; FIDIC contracts; stochastic optimisation; mixed-integer linear programming.

1 Problem Formulation

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2 Problem Formulation

This section develops the formal problem structure for portfolio-level budget allocation in contracted project portfolios. The formulation proceeds across three levels. **Level 1** (Section ??) treats all contractor performance parameters as known and establishes the deterministic MILP whose optimum constitutes a theoretical upper bound: no implementable policy operating under uncertainty can exceed it. **Level 2** (Section ??) reintroduces

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stochastic efficiency and characterizes the resulting multi-stage stochastic program, establishing that its scenario tree grows exponentially in the number of projects and periods. **Level 3** (Section ??) reformulates the intractable stochastic program as a Constrained Markov Decision Process (CMDP), providing an explicit mapping from LP elements to MDP primitives and yielding a Gymnasium-compatible environment on which the PPO agent of Section ?? is trained. The three levels are a coherent arc — Level 1 defines the ceiling, Level 2 explains why it is unreachable, Level 3 provides the tractable path — all sharing the primitives defined next.

2.1 Scope, Assumptions, and Problem Primitives

Scope. The problem is operational, not strategic: the portfolio of contracted projects is fixed, commitments are signed, and the decision-maker exercises a single lever — the budget $x_{i,t}$ directed to each active project i at each discrete period t . All outcomes of interest (milestone achievement, cash balance, schedule adherence, termination) flow through this allocation decision. Portfolio selection is treated as exogenous ?.

Structural assumptions. Four choices underlie the entire formulation. (i) *Independent bilateral contracts*: projects interact only through the shared budget constraint; no resource or workforce sharing occurs, consistent with standard EPC contracting practice. (ii) *Budget as the sole control*: the contractor’s internal resource decisions are absorbed into the efficiency parameter $\eta_{i,t}$; the portfolio manager observes only how much physical progress a budget allocation yields, not how it is achieved. (iii) *Discrete monthly periods*: $\mathcal{T} = \{1, \dots, H\}$ reflects the monthly governance cycle of portfolio management (progress reporting, payment certification, performance review). (iv) *FIDIC contract mechanics*: the milestone payment structure, advance payment and recovery, retention, and termination settlement are parameterized consistently with FIDIC Red, Yellow, and Silver Book forms ?; all contract-specific quantities are calibrated parameters, not fixed constants.

2.1.1 Sets, Indices, and the Decision Variable

[leftmargin=2.2cm, labelwidth=2.0cm]Projects $\{1, \dots, n\}$, indexed by i ; all committed at horizon start. Planning periods $\{1, \dots, H\}$, indexed by t (calendar months). Ordered milestones $\{1, \dots, M_i\}$ for project i , indexed by j , with thresholds $\theta_{i,1} < \dots < \theta_{i,M_i} = 1$. Planned execution window $\{s_i, \dots, f_i\}$; anchors S-curve computations. Projects active at period t ; fixed parameter at Level 1, endogenous at Levels 2–3. Milestones of project i certified by period t : $\{j : \theta_{i,j} \leq P_{i,t}^{\text{actual}}\}$. **Decision variable**: budget allocated to project i at period t . The period- t action is $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$. Portfolio cash balance at the start of period t ; B_1 is given.

2.1.2 Project Parameters

Tables ?? and ?? list all project parameters, fixed at commitment. The planned cumulative expenditure fraction follows a normalized S-curve ?? parameterized by (a_i, b_i) :

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c) = I_{\tau_{i,t}^c}(a_i, b_i), \quad \tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad \tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\},$$

where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function. The planned period expenditure increment is $\Delta \bar{E}_{i,t} = \text{BAC}_i [F_i(\tau_{i,t}^c) - F_i(\tau_{i,t-1}^c)]$. This profile is a fixed parameter derived at commitment from $(a_i, b_i, s_i, f_i, \text{BAC}_i)$; it is never a decision variable.

Parameter	Symbol	Domain	Definition
Planned start & finish Schedule	s_i, f_i	$\mathbb{Z}^+$	Anchor of execution window $\mathcal{A}_i^{\text{plan}}$ and S-curve.
Planned duration Schedule	D_i^{plan}	$\mathbb{Z}^+$	$f_i - s_i + 1$; normalizes progress against plan.
S-curve shape Schedule	a_i, b_i	$\mathbb{R}^+$	Beta CDF parameters governing the planned expenditure profile.
Milestone threshold Schedule	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to certify milestone j .
Delay buffer Schedule	buffer_i	$\mathbb{Z}_0^+$	Periods beyond f_i before forced termination is assessed.
Contract price Contract	CP_i	$\mathbb{R}^+$	Total client payment; $CP_i = \text{BAC}_i(1 + \pi_i)$.
Profit margin Contract	π_i	$\mathbb{R}_0^+$	Contractual markup on BAC_i .
Advance payment ratio Contract	α_i	$[0, 1]$	$A_i = \alpha_i \cdot CP_i$ disbursed at $t = s_i$; recovered via milestone deductions.
Retention ratio Contract	ρ_i	$[0, 1]$	Fraction withheld per milestone; released as $R_i^{\text{ret}} = \rho_i \cdot CP_i$ at completion.
Milestone weight Contract	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of CP_i at milestone j ; $\sum_j \phi_{i,j} = 1$.

Table 1: Schedule and contract parameters for project $i \in \mathcal{P}$ (fixed at commitment).

Parameter	Symbol	Domain	Definition
Budget at completion Cost	BAC_i	$\mathbb{R}^+$	Total planned contractor expenditure; EVM cost basis.
Discount factor Cost	γ	$(0, 1]$	Per-period discount applied to future cash flows; shared across $\mathcal{P}$.
Initial cash balance Cost	B_1	$\mathbb{R}^+$	Portfolio cash balance at horizon start; initial condition of balance recursion.
Deviation penalty weight Penalty	λ_i	$\mathbb{R}_0^+$	Per-period cost per unit absolute progress deviation; calibrated so cumulative penalty at planned finish under sustained contractual-limit delay equals $\Lambda_i \cdot CP_i$.
Deviation penalty cap Penalty	Λ_i	$\mathbb{R}_0^+$	Maximum cumulative penalty as fraction of CP_i ; applies symmetrically to lag and acceleration.
Termination threshold Termination	κ_i	$(0, 1]$	Fraction of $\Lambda_i \cdot CP_i$ at which project enters termination candidacy.
Minimum allocation Termination	m_i^0	$\mathbb{R}_0^+$	Floor budget to hold project in contractually serviceable state while termination conditions are evaluated; no performance-recovery role.
Confirmation window Termination	k	$\mathbb{Z}^+$	Consecutive periods at floor m_i^0 required to confirm abandonment.

Table 2: Cost, penalty, and termination parameters for project $i \in \mathcal{P}$ (fixed at commitment).

2.1.3 State Variables and Transition Dynamics

Project-level state. For each $i \in \mathcal{P}$, $t \in \mathcal{T}$: $P_{i,t}^{\text{actual}} \in [0, 1]$ (cumulative progress), $E_{i,t} \geq 0$ (cumulative expenditure), $SPI_{i,t} = P_{i,t}^{\text{actual}}/P_{i,t}^{\text{plan}}$, $CPI_{i,t} = P_{i,t}^{\text{actual}} \cdot BAC_i/E_{i,t}$, $\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}$ (signed deviation), $\Delta_{i,t}^{\text{pen}} \geq 0$ (cumulative penalty), and $\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$. The portfolio state at period t is:

$$S_t = \left(\{P_{i,t}^{\text{actual}}, E_{i,t}, SPI_{i,t}, CPI_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}\}_{i \in \mathcal{P}}, B_t \right).$$

Budget-to-progress conversion. Budget $x_{i,t}$ converts to progress through the efficiency $\eta_{i,t}$:

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{BAC_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t}, \quad (1)$$

with $P_{i,0}^{\text{actual}} = E_{i,0} = 0$. The scalar $\eta_{i,t}$ absorbs all sources of conversion uncertainty (productivity, supply chain, weather). At Level 1, $\eta_{i,t}$ is replaced by the known deterministic value $\bar{\eta}_{i,t}$ (equal to unity under nominal plan); at Levels 2–3 it is a stochastic primitive realized only after $x_{i,t}$ is committed.

Contractual cash-flow mechanics. Payment inflows follow FIDIC mechanics ?. At initiation ($t = s_i$), the advance payment $A_i = \alpha_i \cdot CP_i$ is received. Milestone j is certified

at the first t with $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$; the net payment (after retention and advance recovery) is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \phi_{i,j} \cdot A_i. \quad (2)$$

Upon completion ($P_{i,t}^{\text{actual}} = 1$) the accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released together with the final milestone payment. The payment identity $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ holds exactly. Aggregate period inflow is:

$$V_t = \sum_{i,j} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{(i,j) \text{ newly certified at } t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \quad (3)$$

where $\mathcal{P}_t^{\text{term}}$ collects projects newly terminated at t , with settlement R_i^{term} defined in (??) below.

Symmetric running deviation penalty. Schedule adherence is priced at every period:

$$c_{i,t}^{\text{dev}} = \min\{\lambda_i |\delta_{i,t}|, \Lambda_i \cdot CP_i\}, \quad \Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}. \quad (4)$$

The penalty is symmetric ($|\delta_{i,t}|$, not $\max\{0, -\delta_{i,t}\}$): both lag and acceleration are penalized, discouraging budget-pushing that risks quality degradation ?. The cap $\Lambda_i \cdot CP_i$ mirrors the contractual delay-penalty ceiling under FIDIC terms.

Termination conditions and settlement. A project transitions to **terminated** at the first period t at which both conditions hold simultaneously:

$$(i) \text{ Penalty saturation: } \Delta_{i,t}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i, \quad (5)$$

$$(ii) \text{ Confirmed floor: } x_{i,\tilde{t}} \leq m_i^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\}. \quad (6)$$

Condition (??) signals exhaustion of contractual schedule tolerance; condition (??) confirms that the portfolio manager has deliberately ceased investment rather than transiently reduced it. Upon termination at t_i^{term} , the net settlement is:

$$R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i(1 - \rho_i) - \left(A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \phi_{i,j} \cdot A_i \right). \quad (7)$$

Negative R_i^{term} (outstanding advance exceeds earned value) constitutes a net obligation recovered from portfolio inflows, consistent with FIDIC termination clauses.

Portfolio cash balance recursion.

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad B_t \geq 0 \quad \forall t \in \mathcal{T}. \quad (8)$$

The constraint $B_t \geq 0$ prohibits borrowing; the portfolio operates under a fixed capital commitment ?. The shared balance B_t is the sole coupling across otherwise independent project dynamics, and its linear scaling with n characterizes the complexity that motivates the intractability argument in Level 2.

2.1.4 Shared Objective

The decision-maker maximizes discounted net cash flow over the horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_{i \in \mathcal{P}_t^{\text{active}}} c_{i,t}^{\text{dev}} \right) \right], \quad (9)$$

subject to, for all $i \in \mathcal{P}$, $t \in \mathcal{T}$:

$$x_{i,t} \geq 0, \quad (10)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad (11)$$

$$x_{i,t} \geq m_i^0 \cdot \mathbf{1}\{\text{status}_{i,t} = \text{active}\}. \quad (12)$$

There is no terminal sum. Completion incentives are embedded structurally: R_i^{ret} and R_{i,M_i}^{net} enter V_t at the certification period; R_i^{term} (possibly negative) enters at termination, creating a natural deterrent against premature abandonment. Constraint (??) ensures each active project remains in a contractually serviceable state; it is vacuous once $\text{status}_{i,t} \neq \text{active}$. The expectation $\mathbb{E}[\cdot]$ collapses to a deterministic evaluation at Level 1 and is retained at Levels 2 and 3.

2.2 Level 1: Deterministic Full-Foresight MILP

Setting $\eta_{i,t} = \bar{\eta}_{i,t}$ (known) renders S_t a deterministic function of $\mathbf{x}_t$, so $\mathcal{P}_t^{\text{active}}$, milestone certification times, and termination periods t_i^{term} are all computable ex ante. Introducing the certification binary $u_{i,j,t} \in \{0,1\}$ ($= 1$ iff milestone j of project i certified by period t) and activity indicator $\text{act}_{i,t} \in \{0,1\}$ (fixed parameter), the stochastic objective (??) collapses to the MILP:

$$\max \sum_{t=1}^H \gamma^{t-1} \left(V_t^{\text{det}} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} c_{i,t}^{\text{dev}} \right) \quad (13a)$$

$$\text{s.t. } B_{t+1} = B_t - \sum_i \text{act}_{i,t} x_{i,t} + V_t^{\text{det}}, \quad B_t \geq 0, \quad \forall t \quad (13b)$$

$$\sum_i \text{act}_{i,t} x_{i,t} \leq B_t, \quad \forall t \quad (13c)$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t} \frac{x_{i,t}}{\text{BAC}_i}, \quad P_{i,t}^{\text{actual}} \leq 1, \quad \forall i, t \quad (13d)$$

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j} u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \quad \forall i, j, t \quad (13e)$$

$$P_{i,t}^{\text{actual}} \leq \theta_{i,j} + \mathbf{M} u_{i,j,t}, \quad u_{i,j,t} \geq u_{i,j,t-1}, \quad u_{i,j+1,t} \leq u_{i,j,t}, \quad \forall i, j, t \quad (13f)$$

$$\begin{aligned} V_t^{\text{det}} = & \sum_{i,j} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}) + \sum_i R_i^{\text{ret}} (u_{i,M_i,t} - u_{i,M_i,t-1}) \\ & + \sum_i A_i \cdot \mathbf{1}\{t=s_i\} + \sum_{i \in \mathcal{P}^{\text{term}}} R_i^{\text{term}} \cdot \mathbf{1}\{t=t_i^{\text{term}}\}, \quad \forall t \end{aligned} \quad (13g)$$

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}} = d_{i,t}^+ - d_{i,t}^-, \quad d_{i,t}^+, d_{i,t}^- \geq 0, \quad \forall i, t \quad (13h)$$

$$c_{i,t}^{\text{dev}} \geq \lambda_i (d_{i,t}^+ + d_{i,t}^-), \quad c_{i,t}^{\text{dev}} \leq \Lambda_i \cdot CP_i, \quad c_{i,t}^{\text{dev}} \geq 0, \quad \forall i, t \quad (13i)$$

$$x_{i,t} \geq m_i^0 \cdot \text{act}_{i,t}, \quad \forall i, t \quad (13j)$$

$$x_{i,t} \geq 0, \quad u_{i,j,t} \in \{0,1\}, \quad u_{i,j,0} = P_{i,0}^{\text{actual}} = 0. \quad (13k)$$

The big- M constant in (??)–(??) is $\mathbf{M} = \max_{i,j} \theta_{i,j}$; tighter values may be derived project-specifically. Constraints (??)–(??) implement the exact if-and-only-if certification logic with monotone, ordered binary activation. Constraints (??)–(??) linearize the absolute-value and cap in (??): the maximizing objective drives $c_{i,t}^{\text{dev}}$ to the tighter of the two bounds, recovering the exact symmetric penalty. The difference $(u_{i,j,t} - u_{i,j,t-1})$ in (??) equals one in the sole period a milestone is first certified, linearizing the trigger without additional binaries. Termination timing t_i^{term} and the settlement R_i^{term} (??) are pre-computed from $\bar{\eta}_{i,t}$ before solving; the termination conditions (??)–(??) are parameter-verifiable at Level 1 and become binding optimization constraints at Levels 2–3.

Upper bound. Let Z_{L1}^* denote the Level 1 optimum. For any causal policy π feasible under uncertainty, $\sup_{\pi} \mathbb{E}[Z^{\pi}] \leq Z_{\text{L1}}^*$: every feasible causal allocation sequence is also feasible in the Level 1 program solved on the realized trajectory, so the full-foresight optimum dominates in expectation. The gap $Z_{\text{L1}}^* - Z^{\pi}$ quantifies the cost of uncertainty and anchors the empirical comparison in Section ??.